

ISM Note 10

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Eddington Luminosity

The gravitational radius is defined as

$$r_g = \frac{GM}{c^2}$$

The Eddington luminosity can be given by

$$\frac{L_{Edd}}{4\pi r^2} \frac{1}{c} \sigma_T = \frac{GMm_p}{r^2}$$

The RHS is the gravitational force, the first fraction on the LHS is the energy flux, combining with inverse of the speed of light gives the momentum flux.

σ_T is the Thomson scattering cross section, which gives **the effective area of the photon captured by the proton**. It is defined as

$$\sigma_T = \frac{8\pi}{3} r_e^2$$

where r_e is the classical electron radius, defined as

$$r_e = \frac{e^2}{m_e c^2}$$

We can derive the Thomson scattering cross section in the low-energy limit where the electron is scattered by photons, where we have

$$P \sim \frac{e^2 a^2}{c^3}$$

where $a = \frac{eE}{m_e}$ is the acceleration of the electron. The energy flux density is given by

$$S = \frac{c}{4\pi} E^2$$

With the energy equation $P = S\sigma_T$, we can derive the Thomson scattering cross section as given above.

Plugging this back the Eddington luminosity equation, we can derive the Eddington luminosity as

$$L_{Edd} = \frac{4\pi GMm_p c}{\sigma_T} \sim 1.3 \times 10^{38} \left(\frac{M}{M_\odot} \right) \text{ erg/s}$$

If we let $L_{Edd} = \epsilon \frac{GM\dot{M}_{Edd}}{r}$, we can derive the Eddington mass flux as

$$\dot{M}_{Edd} = \frac{4\pi}{\epsilon} \frac{r}{r_g} \frac{GMm_p}{\sigma_T c}$$

From the eddington luminosity equation, we can see that **the balance is independent of the radius**, which means that at all radii the balance will hold once $L \sim L_{Edd}$

Now we can make some physical comments on the Eddington luminosity. The gas will inflow or get accreted onto the black hole due to gravity, and since the gas is ionized in electrons and protons, the main gravitational force is acting on the protons. However, when the gas is getting deeper, its temperature will increase and the free-free emission will become more and more important. This will emit photons outside, and can exert radiation pressure on the gas, especially on the electrons. So there is a battle between gravity and radiation, and when these two forces are balanced, we have a typical luminosity, which is the Eddington luminosity.

Sub-Eddington vs Super-Eddington Accretion

If the luminosity of the accretion flow is less than the Eddington luminosity, we call it **sub-Eddington accretion**. In the same way, if the luminosity of the accretion flow is greater than the Eddington luminosity, we call it **super-Eddington accretion**.

The Eddington luminosity gives the maximum accretion rate of a black hole. And the question is when will the sub-eddington and super-eddington accretion happen?

For the sub-eddington accretion, the answer is quite straightforward. If **there is no enough gas supply**, the accretion rate for the black hole will be low. For example, for the black hole in the edge of a galaxy, the gas supply is quite low, and the accretion rate is expected to be sub-eddington.

For the super-eddington accretion, the answer is a little complicated. One reason for the super-eddington accretion is that the accretion is non-spherical and the gas is multiphase. Note that though Eddington luminosity is the theoretical limit for the luminosity of a black hole, **super-Eddington accretion is very common in our universe**.

Inflow of Galaxy Halo with Cooling

Before we consider the inflow of the black hole, we take a look at the inflow at the galaxy halo scale first, which is in $\sim 10^2$ kpc scale.

The big picture is, the gas will flow along the cosmic large scale structure and accrete onto the galaxy halo while losing its energy through cooling. Then what will happen exactly under this picture?

Virial Theorem

The virial theorem applies to a **stable system** where the particle are bounded by **conservative forces**. The virial theorem is given by

$$2\langle T \rangle + \langle U \rangle = 0$$

where T is the total kinetic energy and U is the total potential energy.

The gas in the galaxy halo is basically CGM and we can apply the virial theorem to it. From the virial equation, we can solve the virial temperature of the gas in the galaxy halo as

$$T_{\text{vir}} \sim \frac{\mu m_p}{2k_B} v_c^2$$

where $v_c = \sqrt{\frac{GM_{\text{halo}}}{R_{\text{vir}}}}$ is the circular velocity of the gas in the galaxy halo, $M_{\text{halo}} = M_{\text{DM}} + M_{\text{baryon}}$ is the total mass of the galaxy halo.

μ is the mean molecular weight, which means the average mass of the particles in the gas in proton mass unit. For a fully ionized hydrogen gas, $\mu = \frac{1}{2}$ because there are two particles (proton and electron) for each hydrogen atom. For a fully ionized gas with 75% hydrogen and 25% helium, $\mu \sim 0.6$.

If the gas is stable and the thermal pressure balances the gravitational force, **then the virial temperature is the temperature of the gas in the galaxy halo.**

Dynamical Timescale

One important timescale is the free-fall timescale, which is given by

$$t_{\text{ff}} \sim \frac{R_{\text{vir}}}{v_c}$$

Another important timescale is the cooling timescale, which is given by

$$t_{\text{cool}} \sim \frac{k_B T_{\text{vir}}}{n \Lambda(T_{\text{vir}})}$$

So we have two scenarios of how the gas accretes onto the galaxy halo.

The first scenario is the **hot-mode accretion**, where

$$t_{\text{cool}} \gg t_{\text{ff}}$$

The gas will flow into the galaxy halo and when $v > c_s$ the shock will happen and can convert the kinetic energy into thermal energy. The gas will be heated to the virial temperature and form a hot atmosphere in the galaxy halo. Therefore, the whole process is called the **"virialization"**.

The gas near galaxy cools down gradually and accretes to galaxy via **thermal instability**. Analytical works show that for the hot-mode accretion to happen, we requires $M_{\text{halo}} \geq 10^{11.5} M_{\odot}$.



Fig 1. Hot-Mode Accretion

The second scenario is the **cold-mode accretion**, where

$$t_{cool} \ll t_{ff}$$

Here a stable shock front cannot form and the cool, filamentary gas feeds the galaxy directly. The "virialization" process does not happen and we can know **the halo does not reach a quasi stable thermal state**.

In early universe, there are many small halos and the cold-mode accretion is dominant.



Fig 2. Cold-Mode Accretion

Note that when some non-thermal pressure exists, the virial theorem can be modified as

$$2\langle T \rangle + \langle (U_{\text{grav}} - U_{\text{eff}}) \rangle = 0$$

where U_{eff} is the effective potential energy contributed by the non-thermal pressure like turbulence pressure and magnetic pressure. This implies that the kinetic energy or thermal energy plus the non-thermal energy will balance the gravitational energy **so the virial temperature will become smaller.**

Shocks

Shocks are discontinuities caused by motions $v > c_s$. The place where the discontinuity occurs is called **the shock front**. The gas already getting shocked is called **downstream**, while the gas that has not been shocked is called **upstream**. There are two frames, the lab frame the shock front frame.

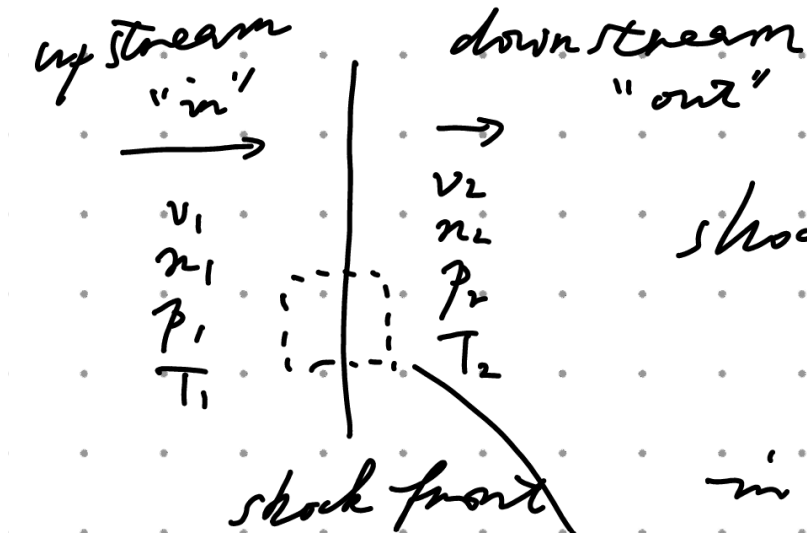


Fig 3. Shock Front Scheme

We consider the shock front frame, and denote the upstream quantities as v_1, n_1, P_1, T_1 and the downstream quantities as v_2, n_2, P_2, T_2 . We have the basic relations

$$v_1 > c_{s,1}$$

$$v_2 < v_1$$

$$n_2 > n_1$$

$$T_2 > T_1$$

And now we are going to ask quantitatively how much the velocity, density and temperature changes across the shock front. This question can be answered by the **shock jump conditions**.

From the mass equation, we have

$$\rho_1 v_1 = \rho_2 v_2$$

From the momentum equation, we have

$$P_1 + \rho_1 v_1^2 = P_2 + \rho_2 v_2^2$$

And we assume the shock is adiabatic which means no thermal energy loss, we have the energy equation

$$\frac{1}{2} v_1^3 + \frac{\gamma}{\gamma - 1} \frac{P_1}{v_1} = \frac{1}{2} v_2^3 + \frac{\gamma}{\gamma - 1} \frac{P_2}{v_2}$$

where γ is the adiabatic index.

We combine these three equations and we can derive the final shock jump conditions as

$$\frac{\rho_1}{\rho_2} = \frac{\gamma - 1}{\gamma + 1} + \frac{2}{(\gamma + 1) \mathcal{M}_1^2}$$

$$\frac{P_2}{P_1} = \frac{2\gamma \mathcal{M}_1^2 - (\gamma - 1)}{\gamma + 1}$$

$$\frac{v_2}{v_1} = \frac{\rho_1}{\rho_2}$$

where $\mathcal{M}_1 = \frac{v_1}{c_{s,1}}$ is the Mach number of the upstream flow.

For a strong shock limit $\mathcal{M}_1 \gg 1$, the shock jump conditions can be simplified as

$$\frac{\rho_1}{\rho_2} = \frac{\gamma - 1}{\gamma + 1}$$

$$\frac{P_2}{P_1} = \frac{2\gamma\mathcal{M}_1^2}{\gamma + 1}$$

$$\frac{v_2}{v_1} = \frac{\gamma - 1}{\gamma + 1}$$

For a monatomic gas, $\gamma = \frac{5}{3}$, we have

$$\frac{\rho_2}{\rho_1} = \frac{v_1}{v_2} = 4$$

and

$$\frac{T_2}{T_1} = \frac{5\mathcal{M}_1^2}{16}$$

This means the downstream gas is much hotter and denser than the upstream gas, means **the upstream kinetic energy is converted into the downstream thermal energy**.

Isothermal Shock

For isothermal shock, the temperature does not change across the shock front, which means $T_2 = T_1$. The shock jump conditions can then be simplified as

$$\frac{\rho_2}{\rho_1} = \frac{v_1}{v_2} = \frac{P_2}{P_1} = \mathcal{M}_1^2$$

Magnetic Shock

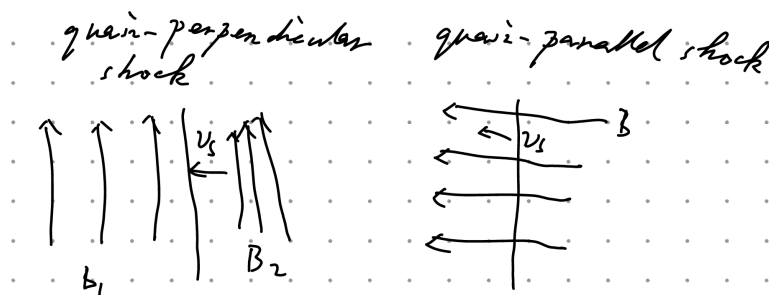


Fig 4. Magnetic Shocks

For the magnetic shock, the magnetic field line can be roughly identified as **quasi-parallel** or **quasi-perpendicular** to the shock front.

For the parallel magnetic shock, the magnetic field does not change across the shock front and the shock jump conditions are the same as the normal shock.

For the perpendicular magnetic shock, due to flux freezing condition, the magnetic field will be compressed across the shock front and the shock jump conditions are different from the normal shock.

Note that inhomogeneities in the upstream flow may lead to baroclinic vorticity or small scale dynamo.

Oblique Shock

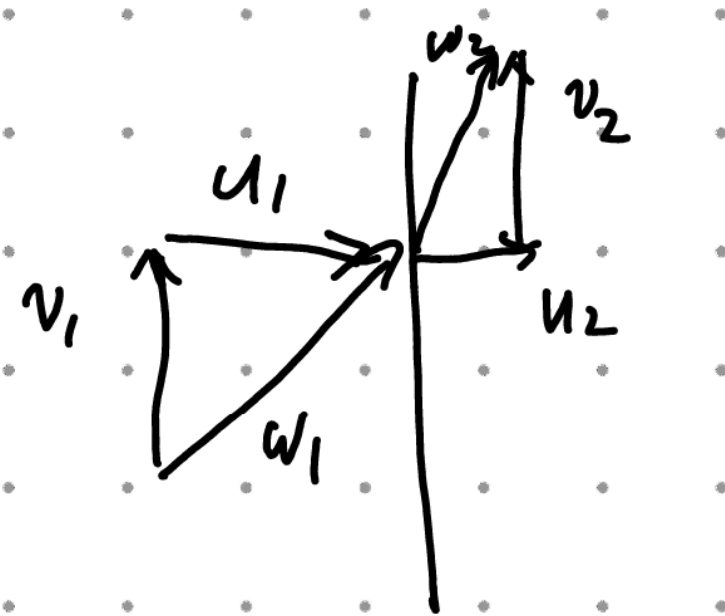


Fig 5. Oblique Shocks

For the oblique shock, the gas flow has a parallel motion to the shock front as well, and due to the shock jump conditions, the velocity of the gas appears to be refracted across the shock front.

Bow Shock

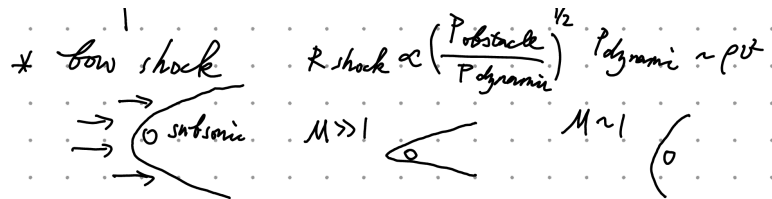


Fig 6. Bow Shocks

For the bow shock, the shock front is curved. The radius or the curvature of the shock front is given by

$$R_{\text{shock}} \propto \left(\frac{P_{\text{obstacle}}}{P_{\text{dynamic}}} \right)^{1/2}$$

where P_{obstacle} is the pressure of the obstacle and $P_{\text{dynamic}} \sim \rho v^2$ is the dynamic pressure of the flow.

Contact Discontinuity

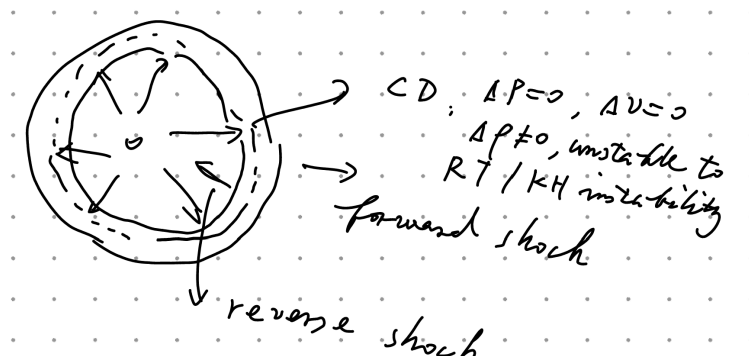


Fig 7. Contact Discontinuity

Usually for the shock we also have the **contact discontinuity**.

For example, for the shock of the supernova explosion, we have the forward shock and the reverse shock. The forward shock is the shock front that propagates into the ISM, while the reverse shock is the shock front that propagates back into the supernova ejecta. The contact discontinuity is the boundary between the forward shock and the reverse shock, where the density and temperature can be discontinuous but the **pressure and velocity are continuous**.

The contact discontinuity(CD) is the interface between the ejecta and the ISM. That's the reason why we care about CD so much, since it is the place to distinguish SN ejecta and ISM.

CD is unstable to fluid instabilities like Rayleigh-Taylor instability and can thus leads to mixture. This consists of one important reason why the crab nebula can have the "finger like" structure.

Riemann Problem

We have the conservation equation for a quantity u as

$$\frac{\partial u}{\partial t} + \nabla \cdot F(u) = 0$$

And we have the initial condition as

$$u(x, t = 0) = \begin{cases} u_L & x < 0 \\ u_R & x \geq 0 \end{cases}$$

Then what is the solution for $u(x, t)$ at $t > 0$? This is the Riemann problem. And we will discuss it further in the next lecture.